

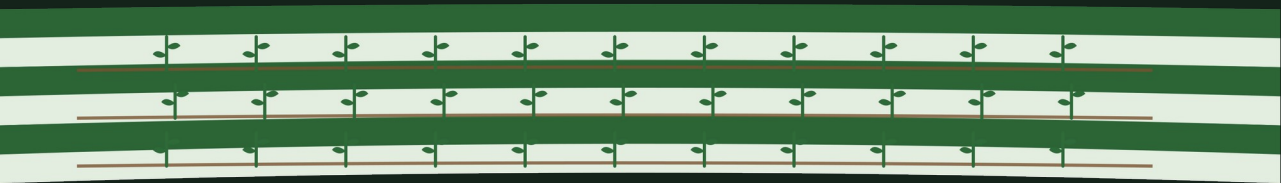
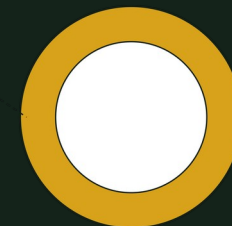
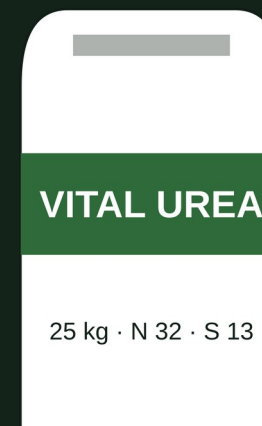


DEALER TRAINING · SEPTEMBER 2026

Vital Urea

Sulfur coated urea. N 32%, S 13%, 25 kg. Pakistan Patent 144684. Made in Lahore.

For VAN dealers and the field team, written so a dealer can repeat it to a grower. Lahore.



What a dealer says in 30 seconds.

Vital Urea is urea with a sulfur coat: 25 kg, N 32% and S 13%, patented and made by VAN in Lahore. Nothing happens to the granule until water reaches it. Then the coating breaks down in steps and the nitrogen comes out over weeks, so the crop meets it while it is still there, and the sulfur goes on with it. In 3 named maize trials, 25 kg of Vital Urea in place of 50 kg of urea at each application gave 5 to 15% more grain with less nitrogen applied. Broadcast, drill or side-dress it, before the water. Never put it in the fertigation tank.

The 4 things it must contain

1

The bag: 25 kg, N 32%, S 13%, patent 144684, PS 217-2023.

2

The mechanism: nothing until water; then released in steps over weeks; the sulfur goes on with the nitrogen.

3

The evidence: 3 named maize trials, 25 kg against 50 kg per application, 5 to 15% more grain.

4

The method: broadcast, drill or side-dress. Never fertigate.

Say it in the grower's own words. Every figure in these 30 seconds is on the bag, in the published crop programme, or in the named evidence on the slides that follow.

THE BAG

What the bag says, and what it is registered as.

25 kg

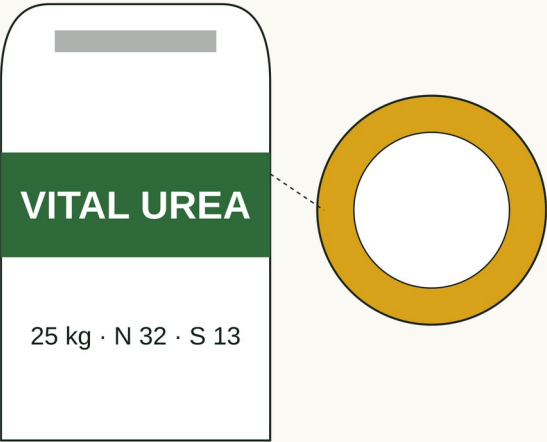
the bag. 8.0 kg of nitrogen and 3.25 kg of sulfur in every bag, from the declared analysis.

N 32%

nitrogen, w/w, declared on the front. The standard's minimum is 32%.

S 13%

sulfur, w/w, fine elemental sulfur on the outside of the granule. The standard's minimum is 13%.



144684

Pakistan Patent, held by VAN, for the sulfur coated urea.

2014

the coating line at the Lahore plant. Every batch is released through LAB 336.

LAB 336

every batch is released through VAN Lab, PNAC accredited, before it ships.

Registered as	Pakistan Standard	Manufacturing licence
Vital Urea	PS 217-2023	CM/L-4126/2025

No expiry date is printed in this deck. The licence number is what a dealer quotes.

A GROWER'S QUESTION

“I put 2 bags of urea on and the field goes yellow again in 3 weeks. Where did it go?”

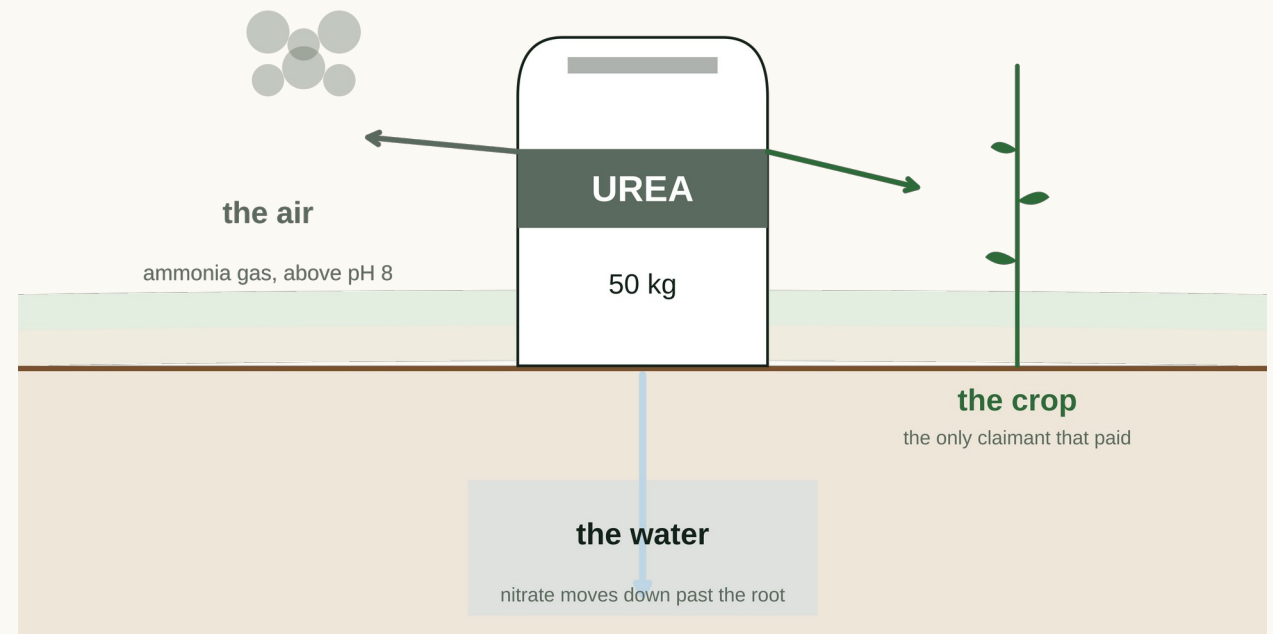
That is the question on the Vital Urea page of van.com.pk, in the grower's own words. A dealer who can answer it does not need a pitch. The answer is 3 ideas: where urea goes, what the coating changes, and what the 3 maize trials measured.

The 3 answers on the next 3 slides, 1 idea each.

A bag of urea has 3 claimants.

The crop, the air and the water. Plain urea dissolves in the first irrigation and everything is exposed at once. Above pH 8 a share leaves as ammonia gas; what dissolves moves down with the water past the root. The farmer paid for all of it.

Say it like this: “Your bag had 3 claimants, and only 1 of them paid for it.”



Drawn from the site's Vital Urea page. No loss fraction is printed, because no Pakistani figure for it is on VAN's record.

The coating limits how much is exposed at once.

Plain urea inside, fine elemental sulfur outside, held by a binder. Nothing happens until water reaches the granule. Then the coating breaks down in steps and the nitrogen is released over weeks, so the crop meets it while it is still there. Soil bacteria turn the fine sulfur into sulfate, and nitrogen needs sulfur to become protein.

Plain urea, after the first water



everything exposed at once, to the crop, the air and the water

Vital Urea, sulfur coated, after the first water



released in steps as the coating breaks down

first water

weeks later

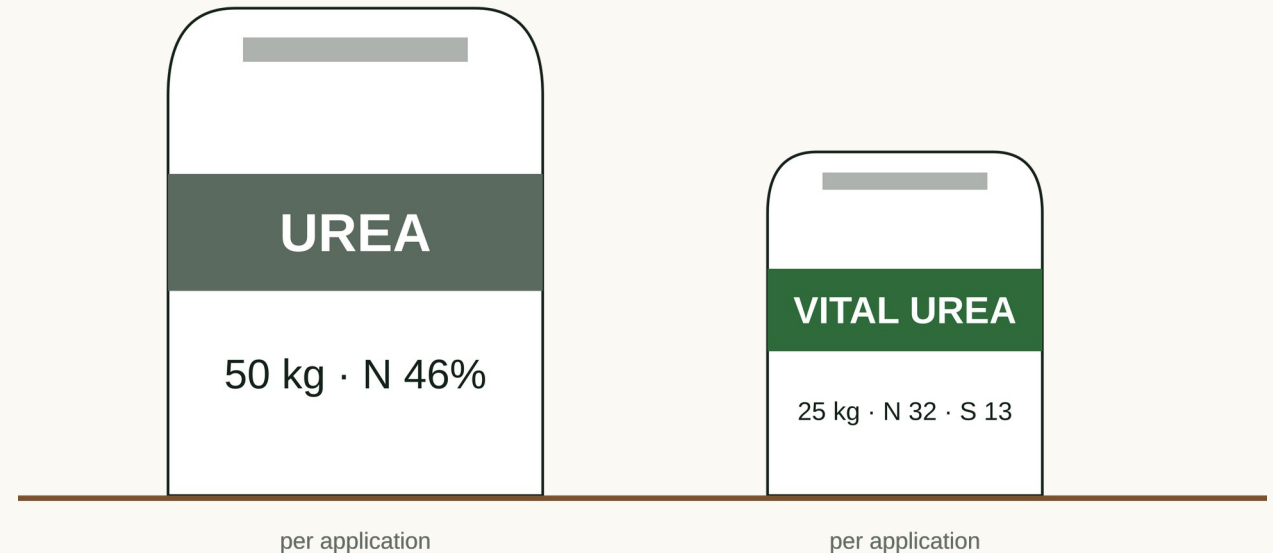
Say it like this: "Nothing happens until the water reaches it. Then it comes out in steps, not all at once."

How it works, not a measurement. No day figures are drawn, because VAN has no laboratory release curve and says so.

Half the weight, and more grain, in 3 maize trials.

In maize trials at Rafhan Maize Products (2022, joint), Descon Research Farm (2024, joint) and Maxim Agri (2025, independent), 25 kg of Vital Urea in place of 50 kg of urea at each application gave 5 to 15% more grain with less nitrogen applied. Do not compare kilograms. Compare what reaches the crop in the weeks it can use it.

Say it like this: “25 kg in place of 50, the same number of times, and the maize gave 5 to 15% more grain.”



Per application. The 3 sites agreed to be named. Maize only; other crops carry no figure.

THE EVIDENCE

The 3 maize trials a dealer may name.

Same design at each site: 25 kg of Vital Urea per application in place of 50 kg of urea per application, the same number of applications.

Site	Year	Kind of trial	Treatment	Result
Rafhan Maize Products	2022	Joint trial	25 kg Vital Urea against 50 kg urea, per application	5 to 15% more grain, less nitrogen applied
Descon Research Farm	2024	Joint trial	25 kg Vital Urea against 50 kg urea, per application	5 to 15% more grain, less nitrogen applied
Maxim Agri	2025	Independent trial	25 kg Vital Urea against 50 kg urea, per application	5 to 15% more grain, less nitrogen applied

The 5 to 15% is the range across the 3 trials, on maize. All 3 sites agreed to be named. No other crop carries a yield figure until its own trial is named. Source: the 27 Sep 2026 ruling and the Vital Urea page of van.com.pk.

What is a standard, what is an observation, and what is neither.

A dealer keeps these 3 apart. The standard is a fact. The observation is a grower’s field, not a trial. Anything else is not said.

Item	What it says	How a dealer may use it
The Pakistan Standard, PS 217	Sulfur coated urea: nitrogen minimum 32%, sulfur minimum 13%, moisture maximum 1.0%, biuret maximum 1.5%.	As fact. The bag is registered against it.
Batch certificates, VAN Lab, March 2024	3 batches (VU24018 to VU24020): N 32.2%, S 13.24%, biuret 0.57 to 0.70%, moisture 0.83 to 0.86%. Nitrogen by Kjeldahl, sulfur gravimetric.	As what a certificate shows. The grower asks for his own batch's certificate.
Khaliqabad, maize, 2023	5 grower sites, August to October, assessed at 12 and 24 days: comparable growth from 25 kg of Vital Urea against 50 kg of urea. No yield was measured.	As a field observation, not a trial. Never as a yield claim.
Other crops	Sugarcane, wheat and other crops have grower observations on file and no named trial.	Not used. No figure for any crop but maize.

Sources: PSQCA standard for sulfur coated urea, Table III; VAN Lab batch certificates SUC/189 to 191, March 2024; the Vital Urea page of van.com.pk for Khaliqabad. All tagged VERIFIED or ON-PACK in the flagship audit, except Khaliqabad, tagged as an observation.

Per application: what the grower buys, in his own back-of-envelope.

Prices from the VAN farmer price sheet, Aug 2026, delivered farmer price. Nitrogen and sulfur from the declared analysis on each bag.

Per application, 1 acre	Bag weight	Nitrogen	Sulfur	Rupees
Urea, 1 bag of 50 kg (N 46%)	50 kg	23.0 kg	0 kg	4,895
Vital Urea, 1 bag of 25 kg (N 32%, S 13%)	25 kg	8.0 kg	3.25 kg	6,500
The difference	25 kg less	15.0 kg less	3.25 kg more	1,605 more

The grower buys 15 kg less nitrogen and 3.25 kg of sulfur, and pays 1,605 rupees more per application. That is the whole of what the counter arithmetic says. What the crop did with it is the trial slide, and it stays on the trial slide.

Urea 50 kg PKR 4,895 and Vital Urea 25 kg PKR 6,500: pricing.ts, the wheat and potato cost plans and FARMER_PRICE. 23.0 kg N = 50 × 0.46; 8.0 kg N = 25 × 0.32; 3.25 kg S = 25 × 0.13.

Per season, at the same number of applications.

The maize programme puts Vital Urea on twice, 1 bag at land preparation and 1 at germination. Wheat is ½ bag. Cotton is 1 bag twice. The table runs 1 to 3 applications so a dealer can read his crop off it.

Applications	Urea kg	Urea N	Urea PKR	Vital Urea kg	N	S	Vital Urea PKR
1	50 kg	23.0 kg	4,895	25 kg	8.0 kg	3.25 kg	6,500
2	100 kg	46.0 kg	9,790	50 kg	16.0 kg	6.5 kg	13,000
3	150 kg	69.0 kg	14,685	75 kg	24.0 kg	9.75 kg	19,500

At 2 applications the grower carries 50 kg instead of 100, buys 30 kg less nitrogen, puts on 6.5 kg of sulfur, and pays 3,210 rupees more for the season. Same number of applications, same stages, same method.

Multiples of the per-application row. Prices: VAN farmer price sheet, Aug 2026. Number of applications per crop: the published crop programme (the programme slides in this deck).



Set the maize result beside the rupees. Do not turn it into rupees.

1,605

rupees more per application for Vital Urea, at the Aug 2026 farmer price

15 kg

less nitrogen bought per application, and 3.25 kg of sulfur on

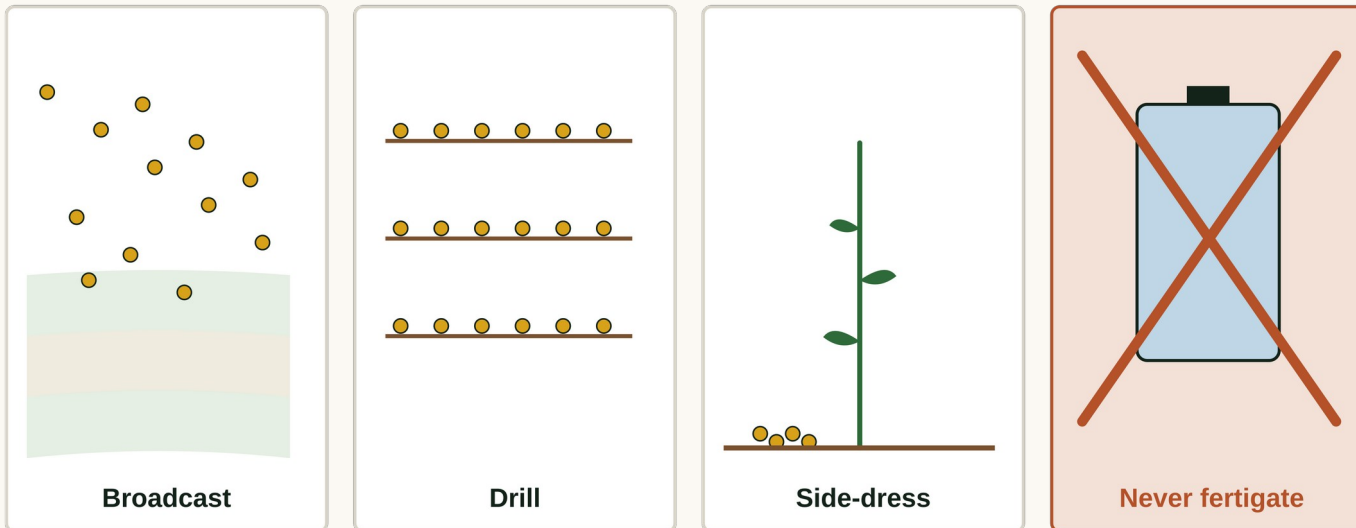
5 to 15%

more maize grain in the 3 named trials, at 25 kg against 50 kg per application

The grower knows his own maize price and his own yield. He does the last step himself. VAN does not print rupees per acre of extra grain, because that number would be VAN's guess about his field, and the trials did not measure his field.

Rupees: pricing.ts. The maize result: the 27 Sep ruling. No yield-value arithmetic, no cost per kg of nitrogen recovered, no nutrient-use-efficiency percentage, by rule.

Broadcast, drill or side-dress. Never fertigate.



Before the water

Apply early, at the stage the programme names, and count from the water. A dry granule does nothing; the first irrigation or rain starts the release.

Whole granules, on the soil

Broadcast at land preparation, drill with the seed row, or side-dress the standing crop. The programme's method column says which.

Never in the tank

A crushed granule is plain urea. A granule in the fertigation tank is a bag of plain urea in the tank: the coating is gone and the 3 claimants are back.

“6,500 for 25 kg, when a 50 kg bag of urea is 4,895?”

The dealer says, out loud

Yes. Per application you buy 25 kg instead of 50, so you are carrying half the weight, and you pay 1,605 rupees more. You are buying 15 kg less nitrogen and 3.25 kg of sulfur, coated onto the same granule. In 3 named maize trials, that 25 kg in place of 50 kg at each application gave 5 to 15% more grain. Work the rupees for your own acre; I will show you the table.

The fact behind it

- Urea 50 kg PKR 4,895; Vital Urea 25 kg PKR 6,500 (VAN farmer price sheet, Aug 2026).
- 23.0 kg N in the urea bag; 8.0 kg N and 3.25 kg S in the Vital Urea bag.
- The maize trials: Rafhan 2022, Descon 2024, Maxim Agri 2025.

Never say: “It pays for itself by X rupees an acre.” VAN prints no rupees-per-acre of extra grain and no cost per kg of nitrogen recovered.

“Less kilograms means less feed. My crop will go hungry.”

The dealer says, out loud

A bag of urea has 3 claimants: the crop, the air and the water. Plain urea dissolves in the first water and everything is exposed at once. The coated granule releases in steps over weeks, so the crop meets the nitrogen while it is still there. That is why 25 kg could give more grain than 50 kg in the maize trials: not more nitrogen, but nitrogen that reached the crop.

The fact behind it

- Flagship answer 1 and 2 on van.com.pk: the 3 claimants; the coating limits how much is exposed at once.
- Trials: 25 kg per application against 50 kg, same number of applications, 5 to 15% more maize grain.

Never say: “Half of urea is wasted” or any loss percentage. No Pakistani loss fraction is on VAN's record, so none is printed.

“I used it and my soil test did not change.”

The dealer says, out loud

It will not, and VAN does not say it will. The acid effect is a zone around each granule for the weeks it is releasing, a few centimetres, not the field. A soil test measures the field. What Vital Urea changes is what the crop meets in those weeks, and that shows in the crop, not in the soil report.

The fact behind it

- Will-not-claim list on the Vital Urea page: no soil pH or soil test change.
- The coating does not change the soil; it limits what is exposed at once.

Never say: “It improves your soil” or “it fixes soil pH”.

“Can I put it through the drip or the flood water?”

The dealer says, out loud

No. Never. Vital Urea is broadcast, drilled or side-dressed as whole granules on the soil. A crushed granule is plain urea, and a granule in the tank is a bag of plain urea in the tank: the coating is gone and the 3 claimants are back. If the grower fertigates his nitrogen, this is not the product for that application.

The fact behind it

- Method on the site and the crop programmes: broadcasting, placement, side dressing. No fertigation row exists for Vital Urea.
- Owner ruling: never fertigate; a crushed granule is plain urea.

Never say: “Dissolve it first” or “it works in the tank if you stir it”.

“Which stage do I put it on?”

The dealer says, out loud

Early, before the water, at the stage the programme names for your crop.
Wheat: ½ bag at land preparation. Maize: 1 bag at land preparation and 1 at germination. Cotton: 1 bag at germination and 1 at grand growth. Rice: 1 bag at early growth. Then count from the first water: nothing happens until it reaches the granule.

The fact behind it

- CROP_PLANS rows for Vital Urea, 12 crops on the programme slides in this deck.
- The release picture is qualitative: no day figures, by ruling.

Never say: A day-count for the release, such as “it lasts 45 days”. VAN has no laboratory release curve and prints none.

“How many bags an acre?”

The dealer says, out loud

What the programme says for your crop, 25 kg in place of each 50 kg of urea at that application. Wheat is $\frac{1}{2}$ bag; maize, cotton and potato are 1 bag at each of 2 stages; sugarcane is 1 bag plus $\frac{1}{2}$ bag in March and again in April. More bags than the programme is not more yield; it is more nitrogen than the crop can meet.

The fact behind it

- Rates per acre as published in the crop programmes (programme slides).
- The trial design was 25 kg in place of 50 kg per application, not extra bags.

Never say: “Double the bags for double the yield.”

“Is it registered? Who says it is what the bag says?”

The dealer says, out loud

It is registered against Pakistan Standard PS 217-2023 under manufacturing licence CM/L-4126/2025, and VAN holds Pakistan Patent 144684 for it. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. Read the batch number off your bag and VAN sends you that batch's certificate, free.

The fact behind it

- Regulatory table on van.com.pk: Vital Urea, PS 217-2023, CM/L-4126/2025.
- Older bags print PS 217-2017 (3rd Rev) on the back; the certificate governs.
- Standard minimums: N 32%, S 13%.

Never say: An expiry date for the licence, or “government approved” as a claim about performance.

“My crop is already yellow. Will this green it up this week?”

The dealer says, out loud

Vital Urea is applied early, at the programme stage, and counted from the water; it releases in steps, not in a rush. So put it on now at the next application the programme names, before the water, and expect the steps, not a jump. It is built for the weeks ahead, and the coating is the reason.

The fact behind it

- Flagship answer 2: nothing until water; then released in steps over weeks.
- Apply early: the audit's method note.

Never say: “It will green up in 3 days.” No day figures, no rescue claim.

Vital Urea in the published programmes, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Land Preparation	½ bag	Broadcasting	25 kg
Maize (Corn)	Land Preparation	1 bag	Broadcasting	25 kg
Maize (Corn)	Germination	1 bag	Side Dressing/Placement	25 kg
Basmati Rice	Early Growth	1 bag	Side dressing/Broadcasting	25 kg
Hybrid Rice	Early Growth	1 bag	Side dressing/Broadcasting	25 kg
Cotton	Germination	1 bag	Placement/Broadcasting	25 kg
Cotton	Grand Growth	1 bag	Placement/Broadcasting	25 kg
Sugarcane, February planting	Germination	1 bag + ½ bag in March; 1 bag + ½ bag in April (two applications each month)	Placement/Broadcasting	25 kg
Ratoon Sugarcane	Land Preparation	1 bag	Placement	25 kg

Rows as published in van.com.pk's crop programmes (catalogue, CROP_PLANS), 12 of the 28 crops. Vital Urea is never fertigated; every row is broadcast, placement or side dressing. A rate is per acre; the crop page scales it to the field.

Vital Urea in the published programmes, continued.

Crop	Stage	Rate per acre	Method	Pack
Ratoon Sugarcane	Germination	1 bag	Placement	25 kg
Potato	Land Preparation	1 bag	Broadcasting	25 kg
Potato	Germination	1 bag	Side dressing	25 kg
Tomato	Land Preparation/Transplantation	1 bag	Broadcasting	25 kg
Chili	Land Preparation/Transplantation	1 bag	Broadcasting	25 kg
Chili	Germination	1 bag	Side dressing	25 kg
Onion	Early Growth (20 DOT)	1 bag	Broadcasting	25 kg
Citrus, 5 years and over	Land Preparation	2 bag	Broadcasting	25 kg
Citrus, 5 years and over	Germination	1 bag	Side dressing/Broadcasting	25 kg

Rows as published in van.com.pk's crop programmes (catalogue, CROP_PLANS), 12 of the 28 crops. Vital Urea is never fertigated; every row is broadcast, placement or side dressing. A rate is per acre; the crop page scales it to the field.

WHAT VAN WILL NOT CLAIM FOR IT

A dealer who says any of these is speaking for himself, not for VAN.

1 A yield figure beyond the 3 named maize trials: 5 to 15% more grain at half the weight per application. Other crops carry no figure until their own trials are named.

2 A laboratory release curve with days on it. VAN has none, so the picture has no numbers.

3 That the coating stops volatilisation. It limits what is exposed at once; it does not change the soil.

4 A soil pH or soil test change. The acid effect is a zone around each granule for the weeks it is releasing.

5 A ratio of nitrogen saved per kilogram of sulfur. The “12 to 15 kg N per kg S” line is off the site and off this deck.

6 That it unlocks soil phosphorus or zinc, or that every field is short of sulfur.

Every batch is tested. The bag carries the proof.

1 · The batch number

It is printed on the bag. Read it to the grower and show him where it is. Every bag of every batch carries one.

2 · The certificate of analysis

For any batch, free on request. Send the batch number off the bag to kisan@van.com.pk or the helpline, and the certificate for that batch comes back. No sample to post.

3 · The laboratory

VAN Lab, PNAC LAB 336, accredited to ISO/IEC 17025:2017. Every batch is released through it before it ships. The certificate carries the laboratory's number.

What a certificate for Vital Urea shows

Nitrogen (Kjeldahl) and sulfur (gravimetric), with biuret and moisture, against PS 217: N minimum 32%, S minimum 13%, moisture maximum 1.0%, biuret maximum 1.5%. The March 2024 batches on file read N 32.2% and S 13.24%. The bag prints the batch number, the Pakistan Standard and the licence number; older bags print PS 217-2017 (3rd Rev) on the back, and the certificate is what governs.

The lines a dealer says in Urdu.

The first line is the promise word from the VAN brand master. The other 3 are the deck's key sentences, translated in a first pass and not yet reviewed by VAN. Say them, and send corrections to partner@van.com.pk.

RELEASE. The promise word for nitrogen.

Urdu first pass, unreviewed

قطرہ قطرہ نائٹروجن

Nothing happens until the water reaches it.

Urdu first pass, unreviewed

پانی پہنچنے تک دانے میں کچھ نہیں ہوتا

Never put it in the tank. Broadcast, drill or side-dress.

Urdu first pass, unreviewed

ٹینک میں کبھی نہ ڈالیں۔ چھٹا دیں، ڈرل کریں یا سائیڈ ڈریسنگ کریں

Do not count the kilograms. Count what reaches the crop.

Urdu first pass, unreviewed

کلو نہ گنیں، وہ گنیں جو فصل تک پہنچے

Write to the address for what you need.

If you grow

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch, a dealer near you.

If you sell

partner@van.com.pk

Dealers, distributors and the trade. Stock, training, corrections to this deck.

Everything else

info@van.com.pk

Supply, the laboratory, a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore. 1 helpline, 3 addresses. No group broadcasts: a grower who writes gets 1 reply.

Vital Urea: broadcast, drill or side-dress, before the water. Never fertigate.

